Noah Sun

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Education

University of Waterloo – BASc. in Mechatronics Engineering (Candidate) September 2025 – April 2030

Relevant Coursework: Algorithms and Data Structures, Circuits, Digital Logic GPA: 3.92 / 4.00

Experience

Embedded Flight Software Developer – Waterloo Aerial Robotics Group September 2025 – Current

- Led 4 engineers build a 32 MB flash translation layer on an STM32 with CRC32 integrity checks across **8192 records**
- Optimized flash I/O CPU time by **15%** with a DMA-driven, non-blocking state machine fed by async RTOS queues
- Verified the driver with **19 GoogleTest** unit tests against a fake SPI flash, covering round-trip, routing and failure cases

V5RC Team Programming Lead – Checkmate Robotics | Demo Video | GitHub June 2019 – August 2025

- Won Create Award at VEX Worlds 2024 among **82 teams** by leading 6 members in robot design and library architecture
- Qualified to VEX Worlds **5 straight years** by building ArkLib, an RTOS-based C++ motion library **reused on 16 robots**
- Achieved **1"/1° movement precision** across **300+ matches** using PID control and 100 Hz sensor-fused odometry
- Automated ring sorting and jam recovery in a 100 Hz RTOS task, detecting jams within **50 ms** using current sensing

VEX AI Competition Team Embedded Systems Lead – Mi3L Schools May 2024 – June 2024

- Enabled vision-based robot autonomy by building a CRC32-validated C++ serial API for Jetson Nano detections through the Intel RealSense camera, winning the Innovate Award and Skills World Champion at VEX AI Worlds 2024

Forward Deployed Engineer – Ascendance Foundry January 2026 – April 2026

- Cut process errors by **80% (50 to 9 per month)** by building **5 AI workflow automations** with input validation
- Saved **6 hours per week** by automating lead discovery and outreach drafts with Python Playwright and Claude API
- Exposed spending inefficiencies across **50+ projects** by building **2 Azure ETL pipelines** feeding Power BI dashboards

Co-founder and Lead Developer – Elapse | Google Play | App Store | GitHub June 2024 – August 2025

- Reached **3500+ downloads** and a **5.0-star rating** by leading 4 developers to ship a Flutter app for VEX tournaments
- Reduced scouting times by **30%** by syncing team data with Firebase and Cloud Functions and predictive match times

Projects

Bare-Metal UART Command Shell (STM32) | GitHub

- Built a bare-metal command shell for an STM32F446 using only **9.3 KB of flash and 2.4 KB of RAM**, with no heap
- **Reduced flash usage by 32%** by replacing the HAL with register-level drivers for GPIO, USART2, and SysTick
- Verified ring buffer correctness with **12 ASan/UBSan-checked unit tests**, including randomized model-based tests

AutoFleet | GitHub

- Coordinated **2 robots** with **1 collision across 5 trials** by building a central aribter with space-time A* planning
- Enabled wireless robot control at **20 ms latency** by programming an ESP32 WiFi-UART bridge with a TCP server
- Reduced localization error by **30%** by fusing EMA-smoothed AprilTag poses with wheel odometry

Redact | Live Demo | GitHub

- Built a full-stack face-blurring app (FastAPI, React, Docker) combining RetinaFace detection with OpenCV tracking
- Sped up video processing **5x to 4 FPS** with **97% face recall** by pairing RetinaFace every 5th frame with IoU tracking

Technical Skills

Languages: C/C++, Python, Java, JavaScript, TypeScript, Dart, MicroPython
Embedded & Hardware: STM32, ESP32, Jetson, Raspberry Pi, Arduino, RTOS, Bare-Metal, UART, SPI, GPIO, Soldering
Robotics & Controls: ROS 2, PID, Motor Control, Kinematics, Odometry, Sensor Fusion, Localization, LiDAR, AprilTags, A*
Software & AI: OpenCV, TensorFlow, FastAPI, Node.js, Express.js, React Native, Flutter, Matplotlib
Data & Tools: PostgreSQL, MySQL, Firebase, Azure, Pandas, Git, Linux, Docker, GitHub Actions, Foxglove